

RESEARCH MINI-PROJECT · DEEP LEARNING & NEURAL NETS

Catching Fraud Without Labels

A deep autoencoder that learns only normal credit-card behaviour and flags anomalies by reconstruction error

Shreyas Sonwane

Sofia University · MSCS2201 Artificial Intelligence

Instructor: Dr. Aleksandr Petiushko

Fraud is rare, late-labelled, and expensive to miss

The problem. Only 492 of 284,807 transactions are fraud ($\approx 0.17\%$), and labels arrive late. Supervised models starve on positives this rare.

The idea. Train an autoencoder on **normal transactions only**. Fraud reconstructs poorly because the model never learned it — so reconstruction error becomes the anomaly score.

Why it matters. No fraud labels needed at training time, and the threshold can be tuned to the business cost of a missed fraud.

0.718

PR-AUC · $\sim 2\times$ the Isolation Forest baseline (0.378)

168 / 246

test frauds caught

56

false alarms across 28,432 normal transactions

Two lineages meet on this benchmark

The dataset & imbalance

Dal Pozzolo et al. (2015) — the ULB / Worldline team behind this benchmark; calibrating probability under extreme class imbalance.

Classic anomaly detection

Isolation Forest (Liu, Ting & Zhou, 2008); One-Class SVM and Local Outlier Factor — the traditional, non-deep family.

Autoencoders for anomalies

Hawkins et al. (2002) replicator nets; Sakurada & Yairi (2014) AE vs PCA; An & Cho (2015) VAE reconstruction probability.

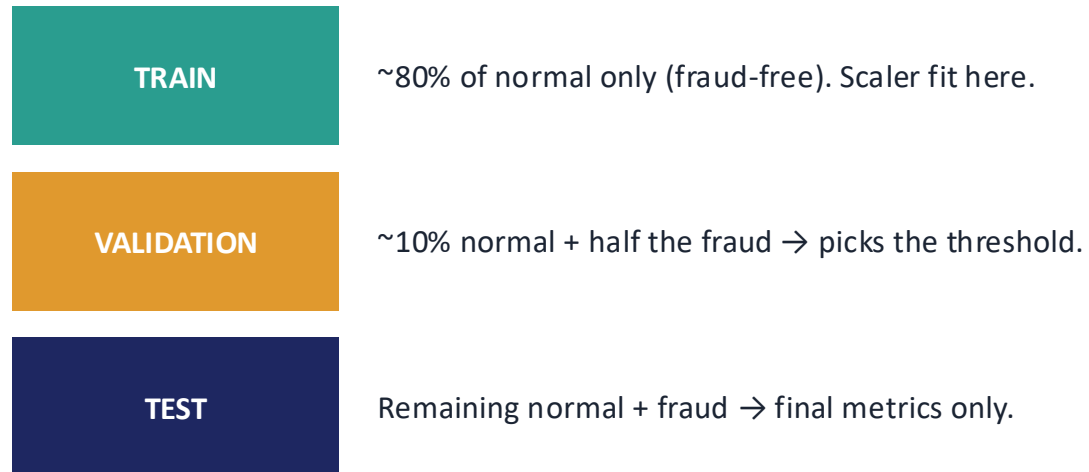
Our framing. Apply the reconstruction-error approach to the standard credit-card benchmark and run it head-to-head against Isolation Forest, with a leakage-safe, validation-chosen threshold.

Citations verified — see README references.

APPROACH

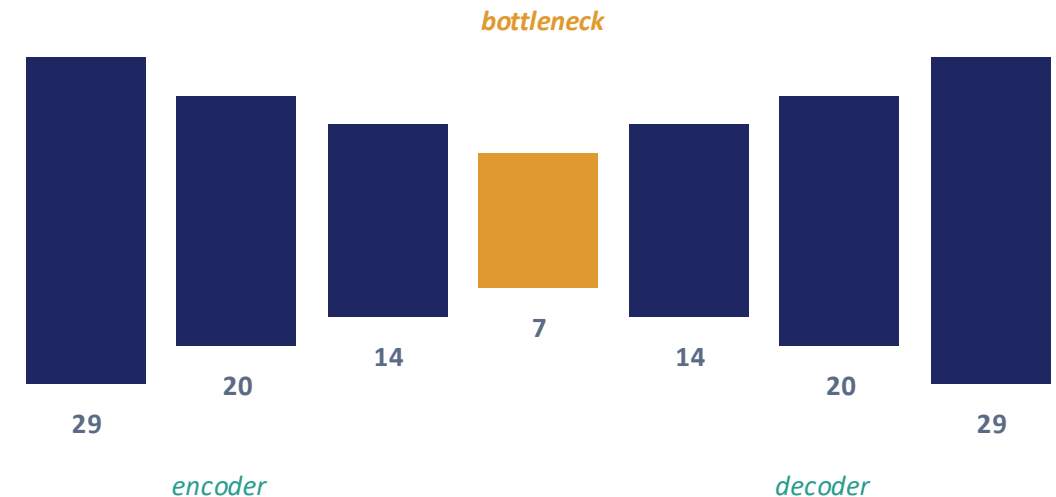
Leakage-safe split, then a small undercomplete autoencoder

Data split — semi-supervised



Drop Time · keep V1–V28 (PCA) + Amount · standardize all · seed 42. **Scaler fit on training-normal only — no leakage.**

Architecture

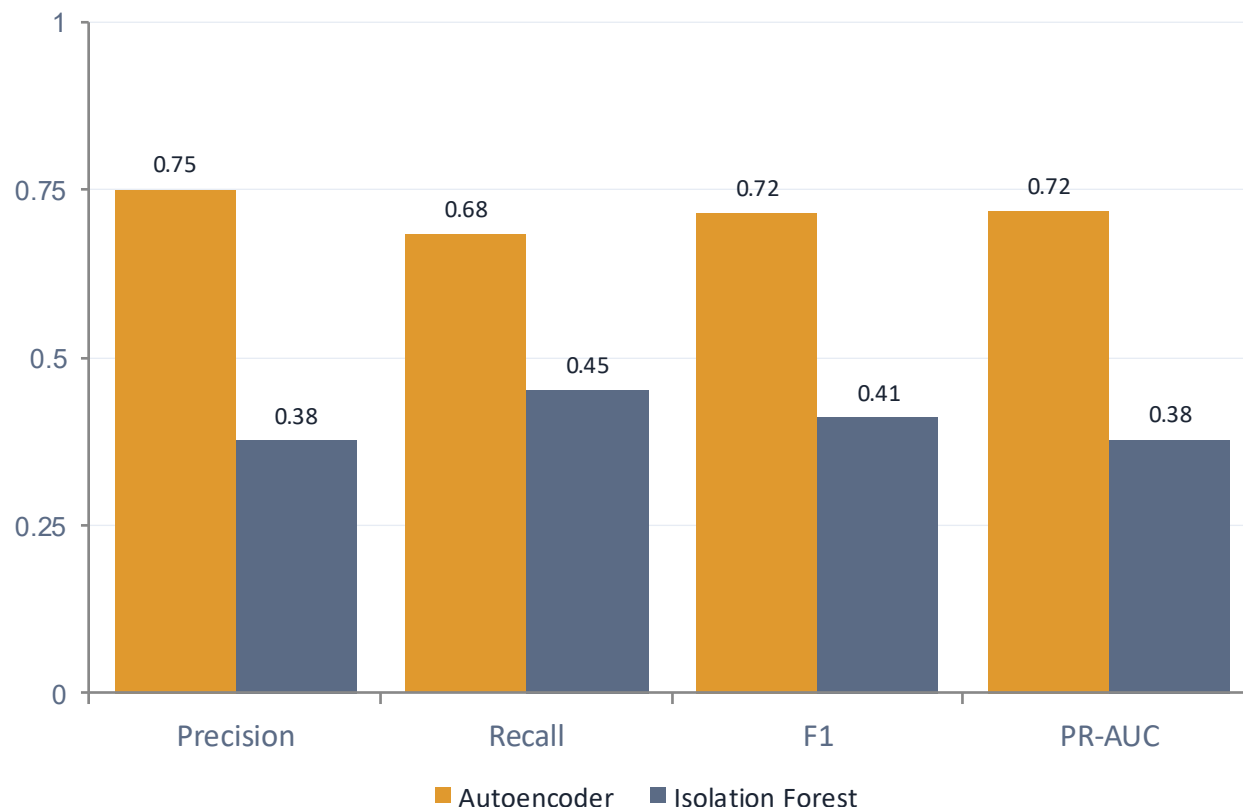


ReLU hidden · linear output · MSE loss · Adam · 40 epochs · batch 256 · ~90s on CPU

Scoring & threshold. Anomaly score = per-sample reconstruction MSE. Threshold = the value that maximizes F1 on validation, then frozen for test.

RESULTS

Beats the classic baseline — and PR-AUC is the honest metric



Why not accuracy?

At 0.17% prevalence, a model that predicts “all normal” scores 99.8% accuracy and catches zero fraud. Accuracy is meaningless here — PR-AUC is the headline.

On the held-out test set

168

frauds caught

78

frauds missed

56

false alarms

28,376

normals cleared

Reproducible, modular, and runnable end to end

GitHub repository

<https://github.com/Shreyas0786/credit-card-fraud-autoencoder>

Run it

```
pip install -r requirements.txt
# place creditcard.csv in data/
python main.py
streamlit run app/demo_app.py
```

Modular

data · model · train · evaluate as separate src/
modules

Reproducible

global seed 42; deterministic split and training

Tested

pytest suite covering the pipeline

DEMO

See it flag fraud in real time

Walkthrough Video

https://youtu.be/MtDNSEQ_yB8

In the Streamlit app: sample a fraud vs a normal transaction, watch the reconstruction error cross (or stay under) the threshold, and read the FLAGGED / NORMAL verdict.